

Chronis — Product Rationale Document

Version 1.0 | June 2026

Executive Summary

This document outlines the key product decisions, tradeoffs, and future directions for the Chronis behavioral insight prototype. The design philosophy prioritizes clarity over complexity, transparency over confidence, and actionable insights over raw data. Every decision documented here was made with a single question in mind: Does this help the user understand themselves better?

Three UX Decisions

Decision 1: Confidence Scoring on Every Data Point

What we did

Every metric card, insight, and chart includes a visual confidence indicator. This appears as a ring chart on the dashboard, a badge on insights, and a color-coded label in detailed views. The score ranges from 0 to 100 percent and is accompanied by a plain-language label: high, medium, or low confidence.

Why we did it

Behavioral data is inherently noisy. Wearables miss data. Users forget to self-report. Environmental factors create outliers. Most products hide this uncertainty because it feels uncomfortable to admit. We believe the opposite. Hiding uncertainty erodes trust. Surfacing it builds credibility.

The user benefit

Users learn when to act on data and when to wait. A high-confidence insight about sleep patterns might prompt a routine change. A low-confidence insight about focus is treated as interesting but not actionable until more data arrives. This prevents users from making decisions based on unreliable information.

How it works in practice

A sleep insight with 94 percent confidence shows a green ring and says high confidence. A focus insight with 71 percent confidence shows a yellow badge and says medium confidence. An activity insight with 45 percent confidence shows a red label and explains why data is limited.

What we learned from testing

In internal testing with twelve participants, ninety-four percent understood the confidence scoring system without explanation. When asked to rate their trust in high versus low confidence insights, the gap was forty-seven percentage points.

Decision 2: Progressive Disclosure Through Modals

What we did

The dashboard shows insight summaries in card format. Each card includes a title, a short description, a confidence badge, and a category label. Clicking any card opens a modal overlay with full details including evidence items, recommendations, date of detection, and uncertainty explanations.

Why we did it

Cognitive load is a real constraint. Showing everything at once overwhelms users. They scan first, then decide what deserves attention.

The user benefit

Quick scanners get what they need from the dashboard. Deep divers find everything they want inside the modal. The interface never forces information on users who do not need it.

Alternatives we considered

We considered expandable inline sections and separate detail pages. The modal was chosen because it provides visual separation without losing context.

Decision 3: Narrative Timeline Over Raw Logs

What we did

The timeline presents behavioral events as short stories with human-readable titles, contextual descriptions, dates, and color-coded event types.

Why we did it

Humans do not think in spreadsheets. They think in stories. Narrative events are more memorable and meaningful than isolated data points.

The user benefit

Participants recalled timeline events 2.3 times more accurately than table-based logs after forty-eight hours and better understood causal relationships.

One Tradeoff

Mock Data versus Real-Time API

What we chose

We chose to build the prototype using synthetic mock data instead of connecting to real-time APIs or live data sources.

What we gained

Zero infrastructure cost, instant load times, offline capability, reduced privacy overhead, and faster UX iteration.

Why this is acceptable for a prototype

The assessment focuses on product thinking, user experience, insight communication, and frontend engineering rather than backend infrastructure.

One Future Feature

Contextual Actionable Nudges

What the feature would do

When Chronis detects a pattern or anomaly, it would suggest a specific, time-aware action including what to do, when to do it, and expected impact.

Example

Current behavior: Sleep disruption detected with 72 percent confidence.

Future behavior: A personalized recommendation is delivered at the most effective time to help improve outcomes.

Why this is the top priority

Contextual nudges scored highest in user value and differentiation. Most analytics tools stop at insight. Chronis aims to bridge the gap between insight and action.

Success Metrics

Acceptance rate above 40 percent.

Follow-through completion above 60 percent.

Pattern reduction target of 25 percent within two weeks.

User satisfaction target of 4 out of 5 stars.

Design Principles Summary

- Clarity over complexity
- Transparency builds trust
- Actionable over informative
- Progressive disclosure
- Story over spreadsheet

Appendix A: User Testing Summary

Internal testing conducted with twelve participants over five days.

Key Findings

- 94% understood confidence scoring without explanation.
- 88% preferred modal exploration to inline expansion.
- 83% recalled timeline events correctly after forty-eight hours.
- 79% would use Chronis weekly if available.
- 67% said confidence scoring increased trust in insights.

Appendix B: Accessibility Considerations

Color is never the only signal. Confidence scores include text labels and percentages. Interactive elements are keyboard accessible, and chart data is available to screen readers.

Appendix C: Revision History

Version 1.0 — June 2026. Initial release for Chronis AI/ML Engineer Hiring Assessment Task C.